Department of Information Engineering and Computer Science

Master's Degree in
Computer Science

Final Dissertation

Maintaining Hitting Sets in Temporal Hypergraphs

Subtitle

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Abstract

1 Introduction

1.1 Context

The increasing availability of relational data that has been studied in recent years has led to a growing interest in the study of data structures capable of representing complex interactions between entities. In the literature, such type of relationship were originally managed by means of *graphs*, which are mathematical structures used to model one-to-one relations between objects. However, many real-world scenarios find this representation too restrictive, when considering interactions between more than just two entities.
Several real-life scenarios cannot be represented without information loss as graphs. Some examples include conversations between multiple people, usually found online in forums, chat or more generally in social media, while others include the interactions between multiple objects or entities, such as those involved in a bank transaction. In these types of contexts, the loss of information when representing these interactions as graphs is unavoidable.
In order to be able to manage this type of data relationship, the concept of *hypergraph* was introduced. Hypergraphs generalize standard graphs by allowing each hyperedge to connect an arbitrary number of nodes, making them suitable for modelling higher-order interactions [1]. This type of data structure is particularly useful when the goal is to represent complex interations without losing structural information, as it is the case with graphs.
In real life scenarios, the interactions between entities are not static, but they evolve over time. New interactions are created, while others tend to disappear, this brings to a complex dynamic system that brings a theoretically static data structure to evolve over time. This type of challenge makes the management of this temporal evolution a necessity, in order to be able to represent the data always in an accurate way. "Just to mention a few examples, temporal graph modelling and analysis of temporal properties can have applications in sociology and social network analysis [...]; security and distributed computing [...]; biology" [2].
The analysis of large dynamic structures also introduces important computational problems [2]. In particular, it is often necessary to continuously update the information of interest without being able to recalculate the entire solution from scratch each time. Consequently, the study of efficient techniques for the maintenance of combinatorial properties and structures in dynamic and temporal environments assumes a central role.

1.2 Problem Statement

When taking in accounting dynamic context

2 Background and Related Work

3 Proposed Approach

3.1 Formal Problem Definition

3.1.1 Introduction

We consider an online variant of the Temporal Hitting Set problem under a sliding window model. To this end, we model the input as a temporal hypergraph $H = (V, E)$, where each hyperedge (S, t) consist of a subset of vertices $S \subseteq V$ and a discrete time step $t \in \mathbb{T}$.

Given a time window of size ϵ, we define the set of active hyperedges at time t as:

$$W(t) = \{(S, t') \in E \mid t - \epsilon \leq t' \leq t\}.$$

At each time step t, the algorithm maintains a hitting set $X(t) \subseteq V$ such that:

$$\forall (S, t') \in W(t), \quad X(t) \cap S \neq \emptyset.$$

3.1.2 Objective

The goal is to maintain a feasible hitting set over time while optimizing the following criteria:

- **Solution size:** minimize $|X(t)|$ at each time step;
- **Update cost:** minimize the number of new nodes taken in consideration when updating the hitting set from $X(t-1)$ to $X(t)$;
- **Computational efficiency:** minimize the time required to update the solution as the sliding window evolves.

Overall, the objective is to maintain a small and stable hitting set while efficiently handling the insertion and expiration of hyperedges induced by the sliding window.

4 Experimental evaluation

5 Conclusions and Future Work

Bibliography

[1] Alessia Antelmi, Gennaro Cordasco, Mirko Polato, Vittorio Scarano, Carmine Spagnuolo, and Dingqi Yang. A survey on hypergraph representation learning. *ACM Computing Surveys*, 56(1):1–38, 6 2023.

[2] Quintino Francesco Lotito and Alberto Montresor. Efficient algorithms to mine maximal span-trusses from temporal graphs. In *Proceedings of the 16th International Workshop on Mining and Learning with Graphs (MLG '20)*, 2020.

Appendix A Attachment

Acknowledgements

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